Assignment 3 – Results
BME.70034: Empirical Asset Pricing

AhRam Cho
20253751

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1 Sample and Empirical Design

The analysis replicates the stock-level return prediction exercise of Gu, Kelly, and Xiu (2020) using monthly U.S. common stock returns from CRSP and firm characteristics from Open Source Asset Pricing. The sample begins in January 1971. The assignment guideline requests results through 2025; however, at the time of data collection, the WRDS CRSP monthly stock file returned observations only through December 2024. Since the prediction target is the next-month return, the final usable feature month is November 2024. I therefore report the second sample as 1971–2024.

The stock universe is restricted to NYSE common stocks, using share codes 10 and 11 and exchange code 1. Firm characteristics are rank-normalized cross-sectionally each month and mapped to the interval $[-1, 1]$. Missing characteristic values are filled with zero, corresponding approximately to the cross-sectional median. The feature set consists of firm characteristics from Open Source Asset Pricing, eight macroeconomic predictors constructed from the Welch and Goyal (2008) dataset (dividend–price ratio, earnings–price ratio, book-to-market, net equity issuance, T-bill rate, term spread, default spread, and stock variance), and the full set of characteristic-by-macro interaction terms, for 836 predictors in total. The choice of characteristics follows the broad anomaly literature (e.g., Fama and French, 1992, 2008; Harvey, Liu, and Zhu, 2016). The prediction target is the next-month stock return, so all characteristics and macro variables dated month t are used to predict stock returns in month $t + 1$.

To keep the rolling-window estimation computationally tractable, a random 30% of NYSE permnos is drawn once at the firm level (with a fixed random seed for reproducibility) and held fixed across all seven models and both sample periods. Because the draw is taken over whole firm histories rather than firm-months, it preserves the cross-sectional and time-series structure of returns and does not introduce bias into return predictability, consistent with the assignment's allowance for random or proportional firm sampling when computational resources are limited. The resulting out-of-sample panel contains roughly 380–390 stocks per month.

Seven models are estimated: OLS+H, OLS-3+H, PCR, ENet+H, Random Forest, NN2, and NN4. OLS-3+H uses the three core characteristics size, book-to-market, and twelve-month momentum, together with macro predictors and their interactions. The other models use the full feature set. Models are estimated with an expanding-window scheme: for each test year, the training sample comprises all months up to three years prior, the two immediately preceding years serve as the validation set for hyperparameter selection, and the test year itself is held out. The out-of-sample evaluation therefore begins in January 2001. Two sets of results are reported: one using the 1971–2016 sample and one using the 1971–2024 sample.

Out-of-sample prediction performance is evaluated using the stock-level out-of-sample R^2:

$$R^2_{OOS} = 1 - \frac{\sum_{i,t}(r_{i,t+1} - \hat{r}_{i,t+1})^2}{\sum_{i,t} r^2_{i,t+1}}.$$

The benchmark is therefore a zero-return forecast.

2 Table 1: Monthly Out-of-Sample Stock-Level Prediction Performance

2.1 1971–2016 Sample

Table 1 reports the monthly out-of-sample stock-level prediction performance for the 1971–2016 sample. The out-of-sample evaluation period runs from January 2001 through December 2016.

Table 1: Monthly out-of-sample stock-level prediction performance, 1971–2016 sample

Model	OOS R^2 (%)	Mean Monthly R^2 (%)	Median Monthly R^2 (%)	Obs.	Months	Start	End
OLS+H	-4.7226	-5.1101	-1.9249	74,752	192	200101	201612
OLS-3+H	-2.3593	-2.0968	-0.3248	74,752	192	200101	201612
PCR	0.5536	0.5547	1.2568	74,752	192	200101	201612
ENet+H	0.3722	-0.0635	0.9415	74,752	192	200101	201612
RF	0.5674	0.1054	0.7859	74,752	192	200101	201612
NN2	-4.5682	-5.2526	-2.3125	74,752	192	200101	201612
NN4	-1.5847	-2.4297	-1.0997	74,752	192	200101	201612

Notes: The table reports stock-level out-of-sample prediction performance for the All sample only. The out-of-sample period is 2001:01–2016:12. The benchmark is a zero-return prediction.

In the 1971–2016 sample, RF, PCR, and ENet+H produce positive full-sample OOS R^2 values. RF has the highest OOS R^2 at 0.5674%, followed closely by PCR at 0.5536%. ENet+H also performs positively, with an OOS R^2 of 0.3722%. In contrast, OLS+H, OLS-3+H, NN2, and NN4 produce negative OOS R^2 values, meaning that their squared prediction errors exceed those of the zero-return benchmark.

The results suggest that some form of regularization, dimension reduction, or nonlinear tree-based modeling is useful for stock-level prediction. The poor performance of OLS+H is consistent with overfitting in a noisy, high-dimensional return-prediction problem. OLS-3+H is more parsimonious but still fails to generate positive OOS explanatory power; this is weaker than its counterpart in Gu, Kelly, and Xiu (2020), where OLS-3+H is marginally positive, and likely reflects the NYSE-only universe and firm subsampling. PCR performs well because it compresses the predictor space into a smaller number of principal components, while ENet+H benefits from shrinkage. RF performs best in this sample, suggesting that nonlinear threshold effects in firm characteristics (Breiman, 2001) may contain useful predictive information.

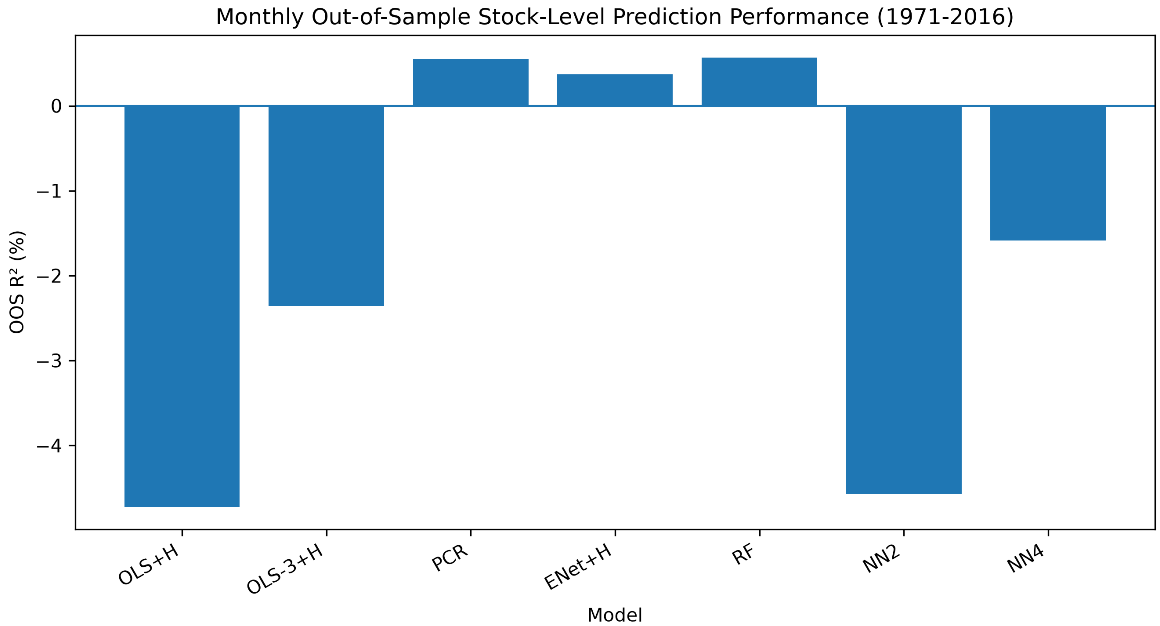

Figure 1: Monthly out-of-sample stock-level prediction performance, 1971–2016 sample

Figure 1 visualizes the same model ranking. RF and PCR are the strongest models, while OLS+H and NN2 are the weakest. The figure makes clear that the flexible neural network models do not necessarily improve point-forecast accuracy in this implementation.

2.2 1971–2024 Sample

Table 2 reports the corresponding results for the extended 1971–2024 sample. The out-of-sample evaluation period runs from January 2001 through November 2024.

Table 2: Monthly out-of-sample stock-level prediction performance, 1971–2024 sample

Model	OOS R^2 (%)	Mean Monthly R^2 (%)	Median Monthly R^2 (%)	Obs.	Months	Start	End
OLS+H	-3.5038	-3.5732	-1.0733	110,028	287	200101	202411
OLS-3+H	-1.8292	-1.3761	0.1119	110,028	287	200101	202411
PCR	0.4662	0.2804	1.1722	110,028	287	200101	202411
ENet+H	0.2680	-0.0201	0.8493	110,028	287	200101	202411
RF	-0.1354	0.0948	0.7114	110,028	287	200101	202411
NN2	-3.7061	-3.8508	-1.3562	110,028	287	200101	202411
NN4	-1.3531	-1.6920	-0.5513	110,028	287	200101	202411

Notes: The table reports stock-level out-of-sample prediction performance for the All sample only. The out-of-sample period is 2001:01–2024:11. The benchmark is a zero-return prediction.

In the extended 1971–2024 sample, PCR and ENet+H remain the only models with positive full-sample OOS R^2. PCR performs best, with an OOS R^2 of 0.4662%, followed by ENet+H at 0.2680%. RF, which performs best in the 1971–2016 sample, becomes slightly negative in the longer sample. OLS+H, OLS-3+H, NN2, and NN4 remain negative.

The extension to 2024 therefore weakens the performance of the more flexible nonlinear models. RF no longer produces a positive full-sample OOS R^2, and the neural networks remain strongly negative. PCR is the most stable model across the two samples. This suggests that dimension reduction is particularly valuable in this implementation, where the signal-to-noise ratio in monthly stock returns is low and the predictor set contains many related characteristics and macro interactions.

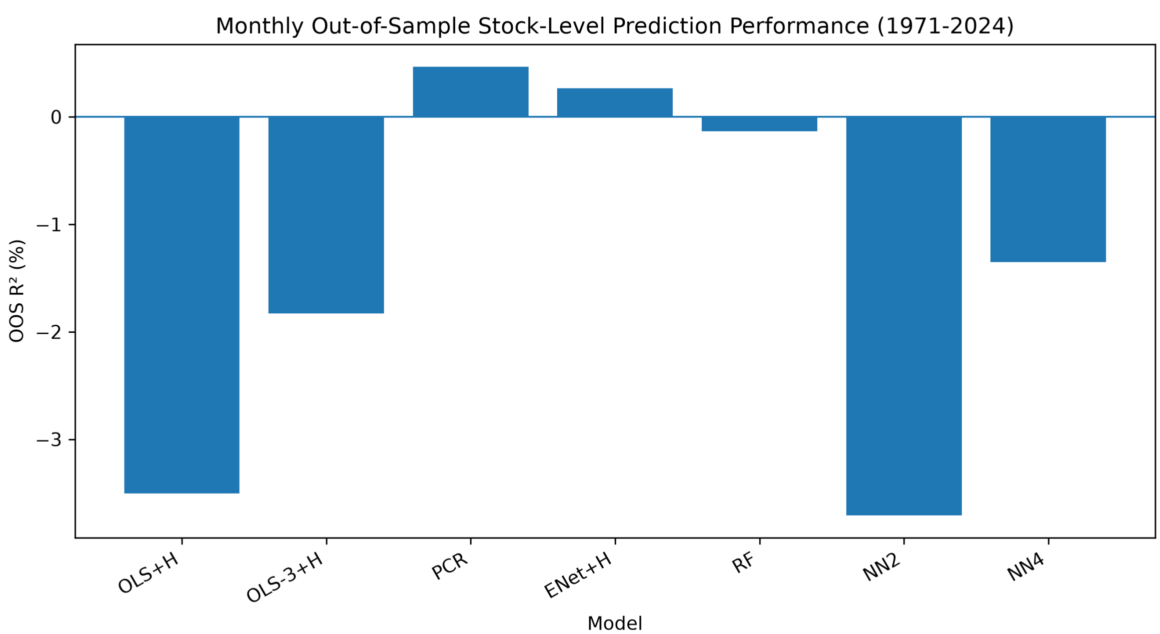

Figure 2: Monthly out-of-sample stock-level prediction performance, 1971–2024 sample

Figure 2 shows the same pattern visually. PCR and ENet+H are the only clearly positive models. RF is close to zero but slightly negative, while OLS+H and NN2 are the weakest models. The difference between the 1971–2016 and 1971–2024 figures suggests that model performance is sensitive to the post-2016 period, especially for RF.

2.3 Comparison with Gu, Kelly, and Xiu (2020)

Relative to the published Table 1 of Gu, Kelly, and Xiu (2020), the penalized and dimension-reduced linear models replicate the paper closely. PCR (about 0.5%) and ENet+H (about 0.3–0.4%) are in line with, or slightly above, the values reported by Gu, Kelly, and Xiu (approximately 0.26% for PCR and 0.11% for ENet+H). The large negative OLS+H R^2 (about -3.5% in the 1971–2024 sample) reproduces their well-known overfitting result (approximately -3.46%), which is a useful validation that the estimation pipeline behaves as intended: an unrestricted high-dimensional linear regression does worse than a zero forecast.

The tree-based and neural-network models underperform the paper, which reports roughly 0.33% for Random Forest and about 0.39% for the neural networks. Here RF is near zero (and slightly negative in the extended sample) and the neural networks are negative. This gap is attributable to three implementation choices. First, the universe is restricted to NYSE stocks, which removes much of the small- and mid-cap cross-section in which the nonlinear models of Gu, Kelly, and Xiu earn a large part of their out-of-sample advantage. Second, only 30% of firms are used. Third, the neural networks are estimated with scikit-learn's multilayer perceptron, which does not implement the batch normalization that Gu, Kelly, and Xiu rely on for stable neural-network training. These differences are expected to compress the advantage of the most flexible models relative to the regularized linear methods.

3 Figure 4: Variable Importance by Model

Variable importance follows Gu, Kelly, and Xiu (2020): for each characteristic, its column—and all of its characteristic-by-macro interaction columns—is set to zero, which, given the cross-sectional rank normalization, is equivalent to setting the characteristic to its median value. The resulting reduction in R^2 is recorded, negative reductions are set to zero, and the values are normalized to importance shares within each model. To avoid re-running the full rolling-window estimation, importance is computed from a single fit per model on the full 1971–E estimation window and measured in-sample over that same window, rather than averaged across all rolling refits as in the original paper. In-sample model reliance is a standard importance notion (it is, for example, what tree-based feature-importance measures report), and measuring it over the multi-decade window—rather than over a short out-of-sample slice—keeps the base R^2 stable and prevents importance from being dominated by corporate events that cluster in any single short period. The reported magnitudes are therefore approximate, but the variable ranking is stable to this choice.

Figure 3 reports variable importance for PCR, ENet+H, RF, NN2, and NN4 in the 1971–2016 sample. The heatmap shows that predictive importance is concentrated in a relatively small set of characteristics. Variables such as dividend omission, short-term reversal, momentum seasonality, cash-flow-related measures, and momentum variables appear among the most important predictors.

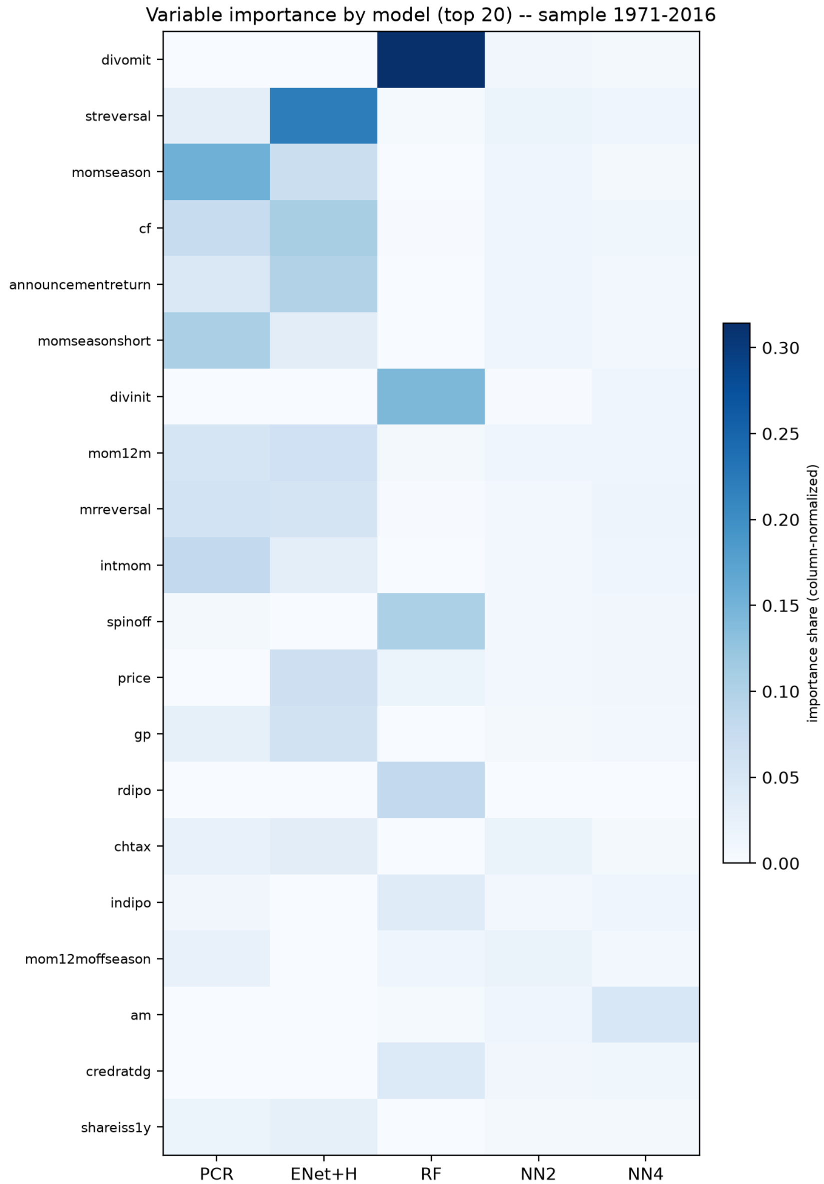

Figure 3: Variable importance by model, 1971–2016 sample

The importance patterns differ substantially across models. RF places particularly high importance on dividend-related variables, especially dividend omission and dividend initiation. This is partly a property of the feature set—the Open Source Asset Pricing library includes sparse, event-type indicators (dividend omission and initiation, recent IPO, spinoff) that the original characteristic set of Gu, Kelly, and Xiu does not emphasize—and partly a known property of impurity-based importance for tree models, which can over-weight sparse signals that the forest memorizes in-sample. This is consistent with RF's near-zero out-of-sample R^2 in Table 2: the in-sample event patterns it relies on do not generalize. ENet+H places relatively more emphasis on short-term reversal and cash-flow-related predictors, consistent with sparse linear shrinkage. PCR assigns importance more broadly across reversal, momentum, and seasonality-related variables, reflecting the fact that principal components summarize common variation across predictors rather than selecting a small number of variables directly.

Figure 4 reports the corresponding variable importance heatmap for the extended 1971–2024 sample.

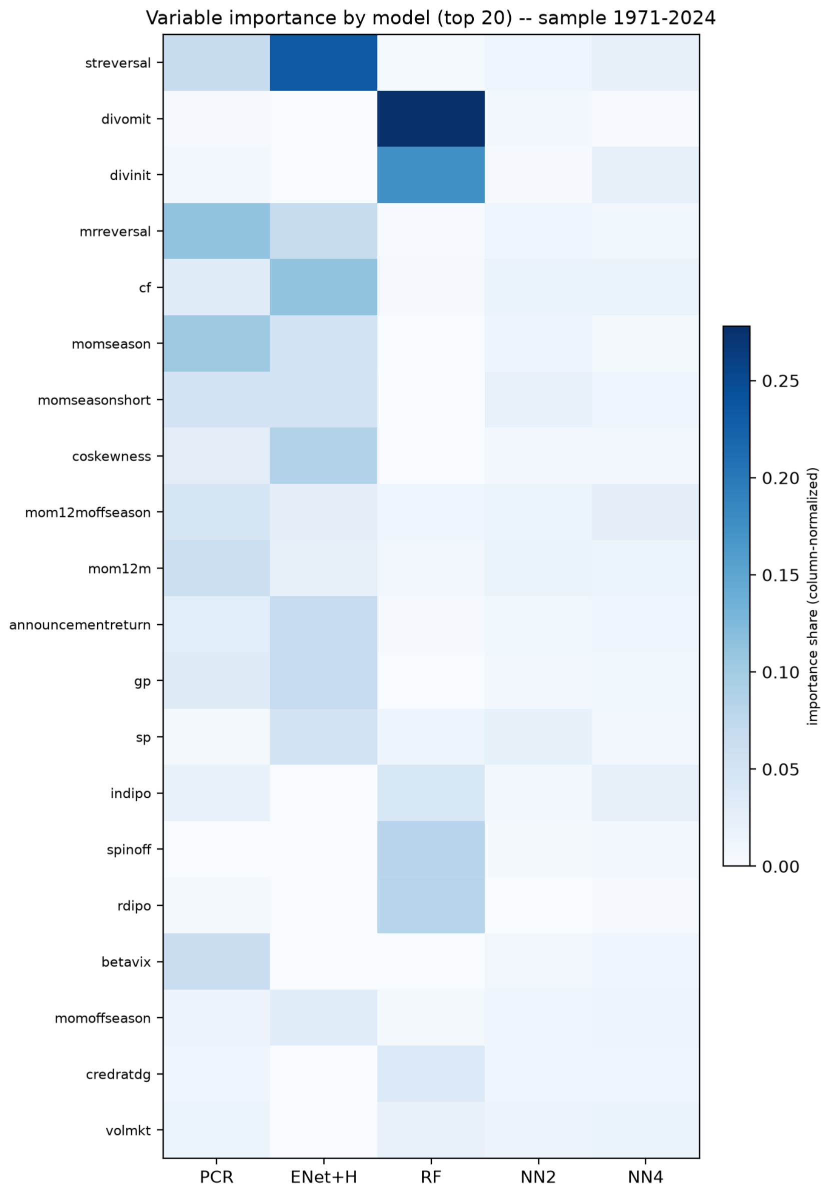

Figure 4: Variable importance by model, 1971–2024 sample

The extended sample shows a broadly similar set of important variables. Short-term reversal, dividend omission, dividend initiation, cash flow, momentum seasonality, and momentum-off-season variables continue to appear near the top of the importance ranking. However, the relative importance of variables changes across models and sample periods. RF remains highly sensitive to dividend-related variables, while ENet+H continues to emphasize reversal and cash-flow-related signals. The neural network models show weaker and more diffuse importance patterns, consistent with their weaker Table 1 prediction performance and with the underfitting discussed above.

Overall, Figure 4 suggests that the most stable predictive signals in this implementation are related to reversal, dividend events, cash flow, and momentum seasonality. These variables remain important across both samples, although the exact ranking varies by model.

4 Figure 9: Cumulative Returns of Machine Learning Portfolios

Figure 5 reports cumulative returns of long-short machine learning portfolios over the 2001–2016 out-of-sample period. Each month, stocks are sorted into deciles by predicted return; the portfolio is long the top decile and short the bottom decile, value-weighted within each leg by market equity, and held for one month. The figure also includes the S&P 500 excess return (S&P 500 return minus the risk-free rate), and shaded areas indicate NBER recession periods.

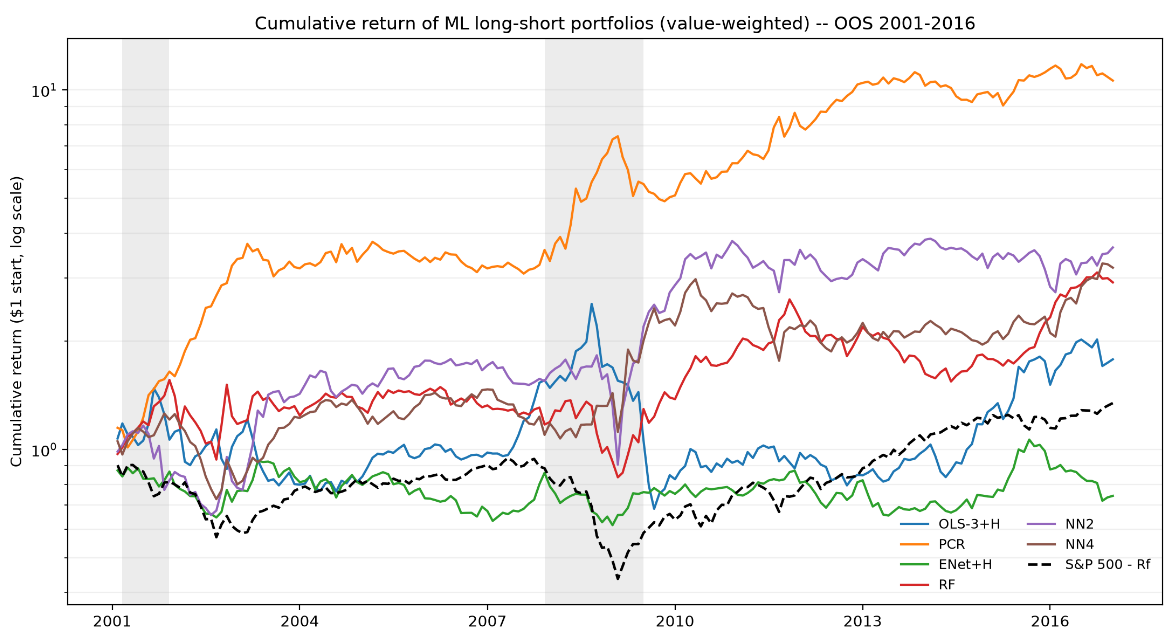

Figure 5: Cumulative return of machine learning portfolios, 2001–2016 out-of-sample period

PCR generates the strongest cumulative portfolio performance in the 2001–2016 period. While its stock-level OOS R^2 is also positive, the magnitude of its portfolio dominance far exceeds its small R^2 edge, underscoring that long-short performance is driven by a model's ability to *rank* stocks cross-sectionally rather than by squared-error accuracy. NN2 and NN4 reinforce this point: they produce positive cumulative returns despite weak point-prediction performance. Indeed, OOS R^2

measures squared forecast error at the individual stock level, while long-short portfolio performance depends mainly on the ordering of predicted returns.

RF, OLS-3+H, and ENet+H produce more moderate cumulative returns. ENet+H is relatively stable but does not compound as strongly as PCR. The S&P 500 excess return benchmark performs well over the sample, but the PCR long-short strategy substantially outperforms it.

Figure 6 extends the cumulative return analysis through 2024.

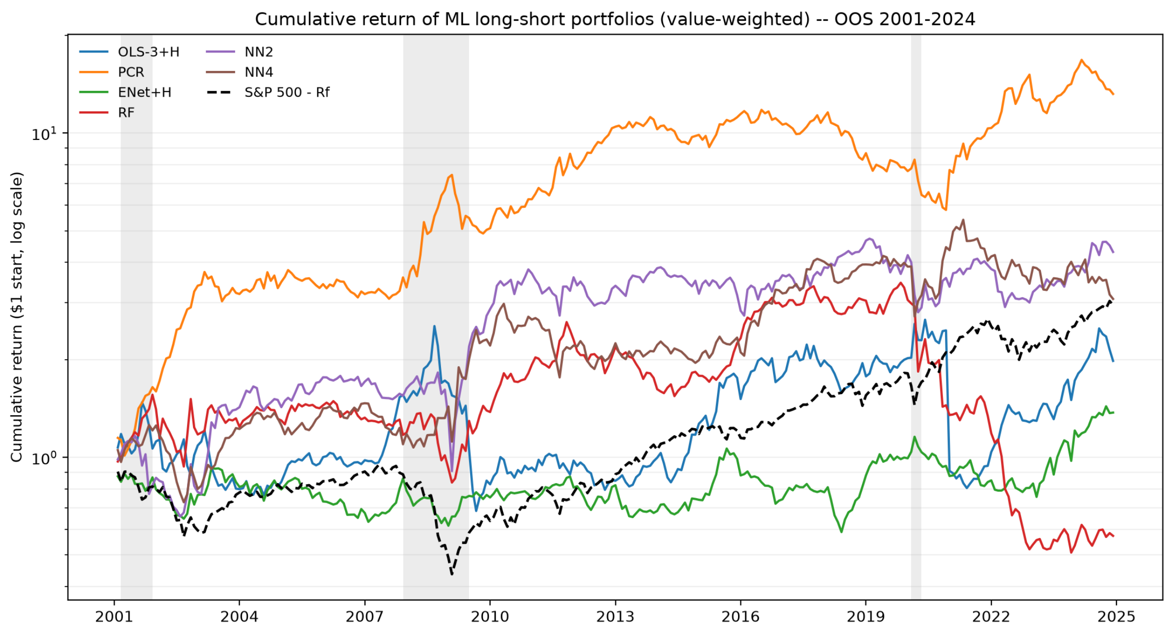

Figure 6: Cumulative return of machine learning portfolios, 2001–2024 out-of-sample period

In the 2001–2024 sample, PCR remains the strongest long-short portfolio. Its cumulative return rises substantially over the full out-of-sample period and remains above the other model portfolios. NN2 and NN4 also generate positive cumulative performance, although they do not match PCR. OLS-3+H performs moderately, while ENet+H produces lower but relatively stable returns.

RF shows a more unstable pattern. It performs reasonably well during parts of the sample but deteriorates sharply after 2021. This is consistent with Table 2, where RF's full-sample OOS R^2 becomes slightly negative in the extended sample. The post-2020 period therefore appears to weaken the RF signal substantially.

The recession shading highlights the behavior of the strategies during stress periods. During the 2008–2009 financial crisis, several strategies experience sharp volatility, but PCR subsequently recovers strongly. During the 2020 recession, model portfolios react differently: OLS-3+H and RF show large movements, while PCR continues to compound strongly after the shock. The post-pandemic period produces substantial divergence across models, with PCR remaining robust and RF deteriorating. Because the portfolios are value-weighted within thin NYSE deciles, individual months can be influenced by a small number of large stocks; the qualitative ranking of the strategies, however, is stable across the sample.

5 Discussion

The results show that monthly stock-level prediction remains difficult. Several models produce negative OOS R^2, meaning that their squared prediction errors are worse than those of a zero-return forecast. This is particularly true for OLS+H and NN2. The poor performance of OLS+H is consistent with overfitting in high-dimensional linear prediction and, as noted above, closely matches the corresponding result in Gu, Kelly, and Xiu (2020). The weak performance of NN2 and NN4 reflects the neural-network implementation: scikit-learn's multilayer perceptron does not include the batch normalization that Gu, Kelly, and Xiu rely on to stabilize training, so the networks tend to produce miscalibrated, upward-biased point forecasts that inflate squared error.

PCR is the most reliable model in this implementation. It has positive OOS R^2 in both samples and generates the strongest long-short portfolio returns. This suggests that dimension reduction is especially useful when predictors are noisy and highly correlated, in the spirit of the dimension-reduction approach of Kelly, Pruitt, and Su (2019). ENet+H also performs reasonably well, indicating that shrinkage improves over unrestricted linear regression.

The comparison between Table 1 and Figure 9 highlights that stock-level prediction accuracy and portfolio performance are related but not identical. PCR performs well on both dimensions, but NN2 and NN4 generate positive long-short returns despite negative OOS R^2. This means that models can have weak point forecasts but still contain useful cross-sectional ranking information. Since long-short portfolio returns depend on ranking rather than exact return forecasts, models with poor squared-error performance can still produce economically meaningful trading strategies.

The variable-importance results indicate that reversal, dividend-related variables, cash-flow measures, and momentum seasonality are among the most important predictors. These variables remain important across both the 1971–2016 and 1971–2024 samples, although the relative weights differ across models. RF is especially sensitive to dividend-related variables, while PCR and ENet+H rely more broadly on reversal, momentum, and cash-flow-related information.

Overall, the results support the central message of the machine-learning asset-pricing literature: predictive information exists in firm characteristics, but extracting it requires regularization, dimension reduction, or nonlinear modeling. In this implementation, the most robust approach is PCR. Highly flexible models such as neural networks and random forests are not uniformly superior, and their performance is sensitive to the sample period and to the implementation choices noted in Section 2.

6 Conclusion

This assignment replicates the main empirical outputs of Gu, Kelly, and Xiu (2020) using CRSP returns, Open Source Asset Pricing firm characteristics, macroeconomic predictors, and machine-learning prediction models. The results are reported for two samples: 1971–2016 and 1971–2024.

The main findings are threefold. First, PCR and ENet+H are the most consistent models in terms of stock-level OOS R^2, remaining positive in both samples, with PCR the most stable;

Random Forest attains the highest OOS R^2 in the 1971–2016 sample but turns negative once the sample is extended to 2024. Second, PCR also generates the strongest cumulative long-short portfolio performance, indicating that its prediction signals are economically meaningful. Third, variable-importance results suggest that reversal, dividend events, cash-flow measures, and momentum seasonality are the most important predictors in this implementation.

The results also show that more flexible models do not automatically dominate simpler regularized methods. Neural networks produce positive long-short portfolio returns but weak stock-level OOS R^2, while RF performance deteriorates in the extended sample. Consistent with the comparison to Gu, Kelly, and Xiu (2020), the shortfall of the tree and neural-network models is plausibly driven by the NYSE-only universe, the 30% firm subsample, and the absence of batch normalization in the neural-network implementation. The evidence therefore favors dimension reduction and shrinkage as robust tools for stock-level return prediction in this replication.

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